



جامعة العلوم الإسلامية العالمية
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كلية تكنولوجيا المعلومات
Faculty of Information Technology

Course: System Analysis and Design (0604334)

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PrintSmart: Unified Print Estimation & Inventory System

Phase 2: Project Planning & System Modeling

1. Executive Summary & Business Context

Building upon the System Proposal (Phase 1), Phase 2 transitions the **PrintSmart** project from problem definition to structural and operational planning. The print shop currently suffers from operational bottlenecks due to manual pricing estimation via static Excel sheets, lack of dynamic B2B/B2C calculations, and unrecorded inventory waste. This phase utilizes project management tools (PERT, Gantt) to establish a realistic development timeline, and structured analysis tools (DFDs, Event Response Table) to map the print shop's complex business logic into a scalable software architecture.

2. Project Management & Scheduling

To ensure the timely delivery of the PrintSmart system, the development process was divided into modular work packages.

2.1. Task Estimation (PERT Analysis)

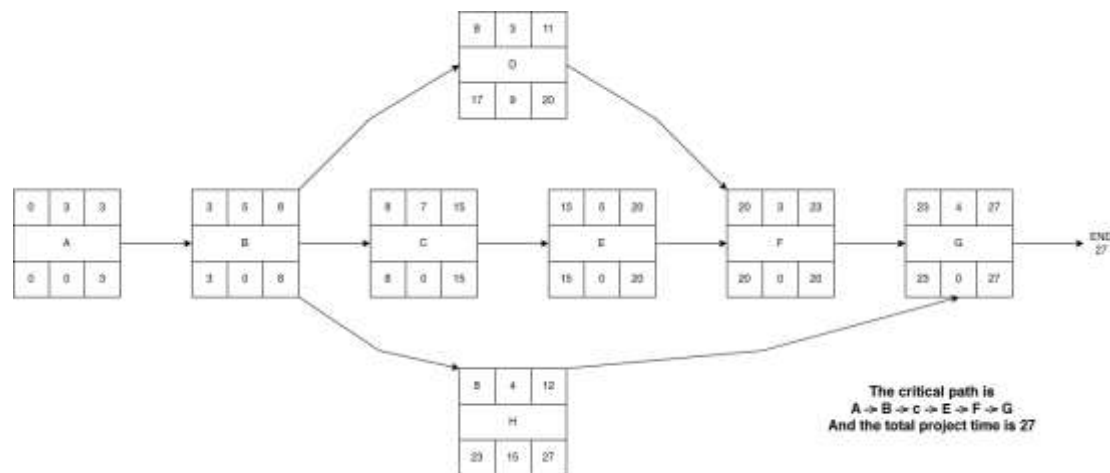
We utilized the Three-Point Estimation technique to accurately calculate the Expected Time TE for each development phase, mitigating risks associated with complex programming modules.

Table 2.1: Project Tasks and Expected Time Calculations

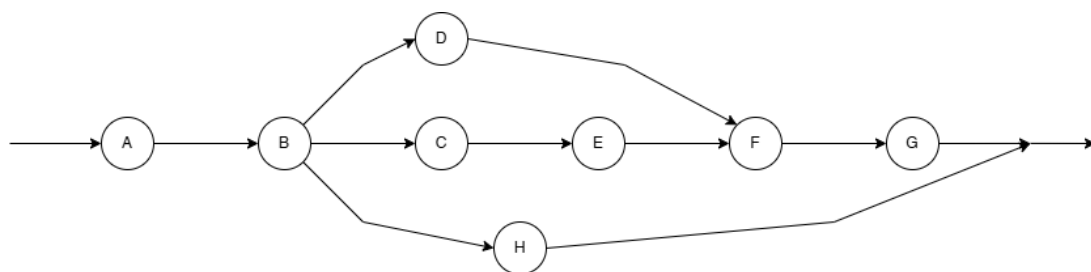
Task Identifier	Task Description	Predecessor Task(s)	Optimistic time (O)	Most Likely time (M)	Pessimistic time (P)	Expected Time (TE)
A	Finalize Project Charter & Scope	None	2	3	4	3
B	Gather Requirements & Define Logic	A	3	5	7	5
C	Design System Architecture & DFDs	B	4	6	14	7
D	Configure Database Tables	B	2	3	4	3
E	Develop Core Programming Modules	C	3	5	7	5
F	System Integration & Testing	D, E	2	3	4	3
G	User Acceptance Testing & Deployment	F	2	4	6	4
H	Write Documentation & Training Manuals	B	2	4	6	4

2.2. Network Diagram & Critical Path

Based on the task dependencies, a Network Diagram was constructed to visualize the workflow and identify the most time-critical sequence of activities.

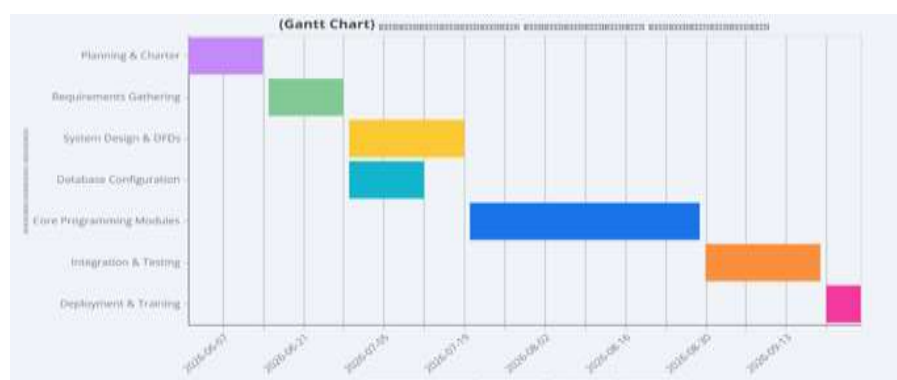


- **Critical Path Identification:** Based on our forward and backward pass calculations, the Critical Path is **A → B → C → E → F → G**.
- **Total Project Duration:** The project will require a minimum of **27 weeks** to complete. Any delay in these critical tasks will delay the entire system's deployment. Tasks D and H possess slack time, allowing flexibility in their execution.



2.3. Gantt Chart

The Gantt chart translates the PERT logic into a calendar-based schedule. It highlights the parallel execution of non-dependent tasks (such as configuring the database while designing the architecture) to optimize the development timeline.

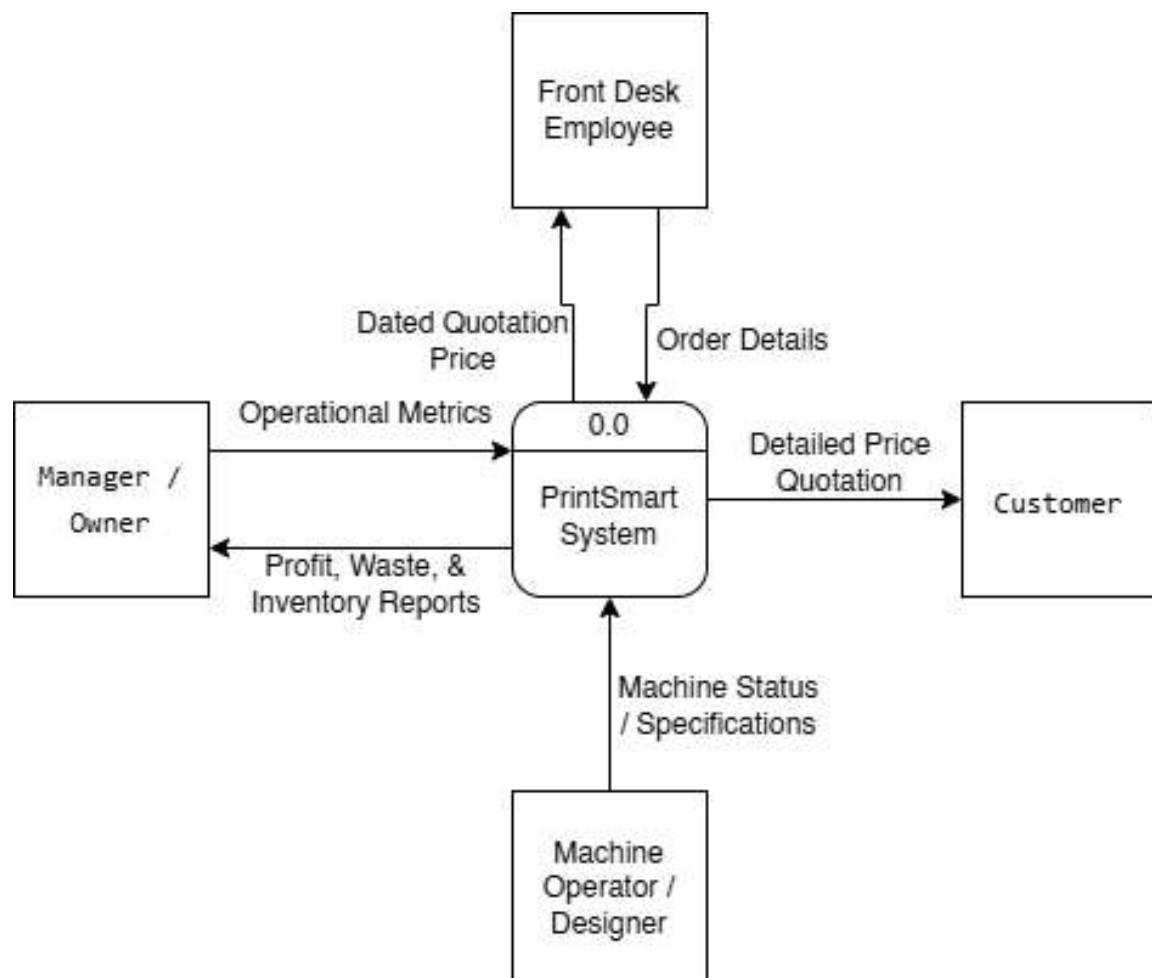


3. System Analysis & Structured Modeling

To digitize the print shop's manual processes securely and efficiently, we modeled the data flow from a high-level business perspective down to the physical database implementation.

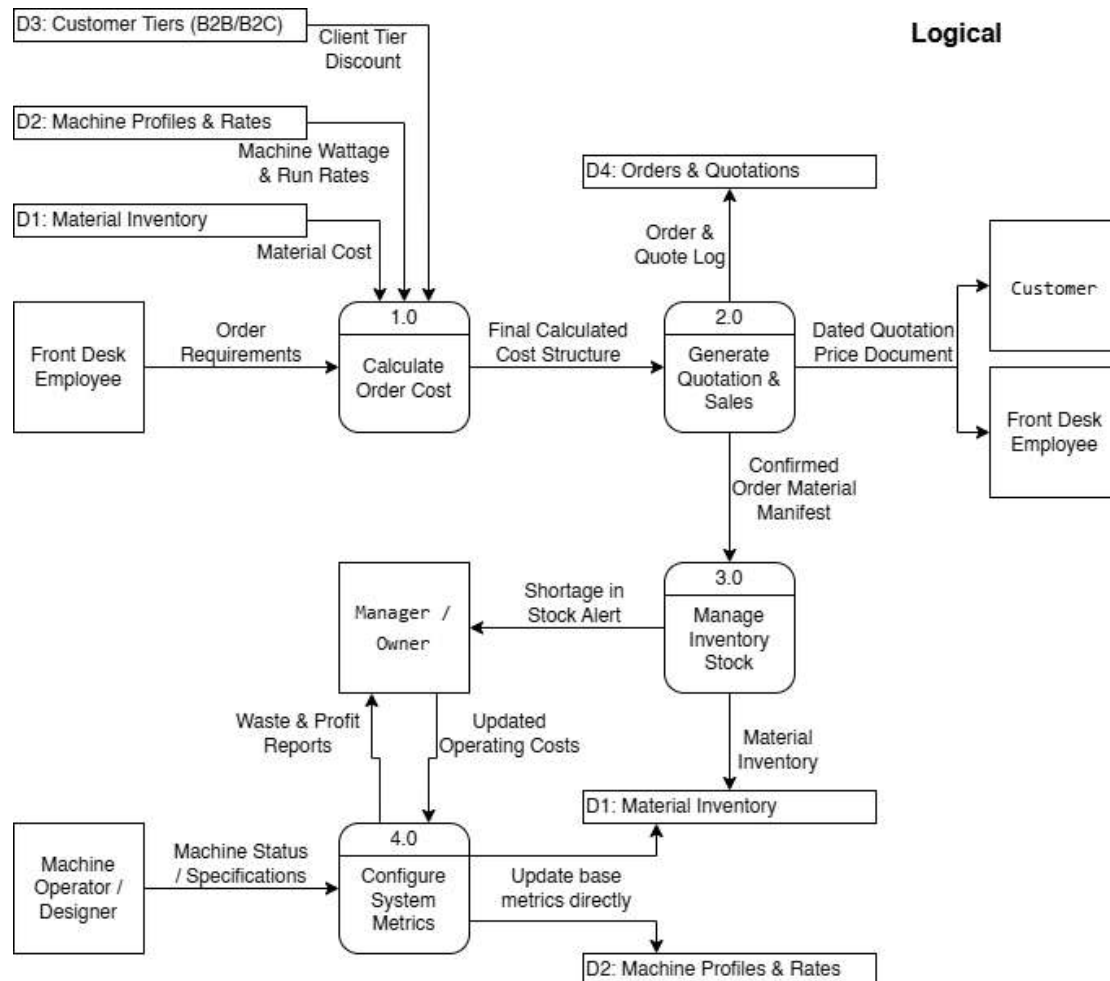
3.1. Context Diagram (Level 0)

The Context Diagram establishes the system boundary. It illustrates how the central PrintSmart System interacts with external entities (Customer, Front Desk Employee, Machine Operator, and Manager/Owner) without revealing internal mechanics.



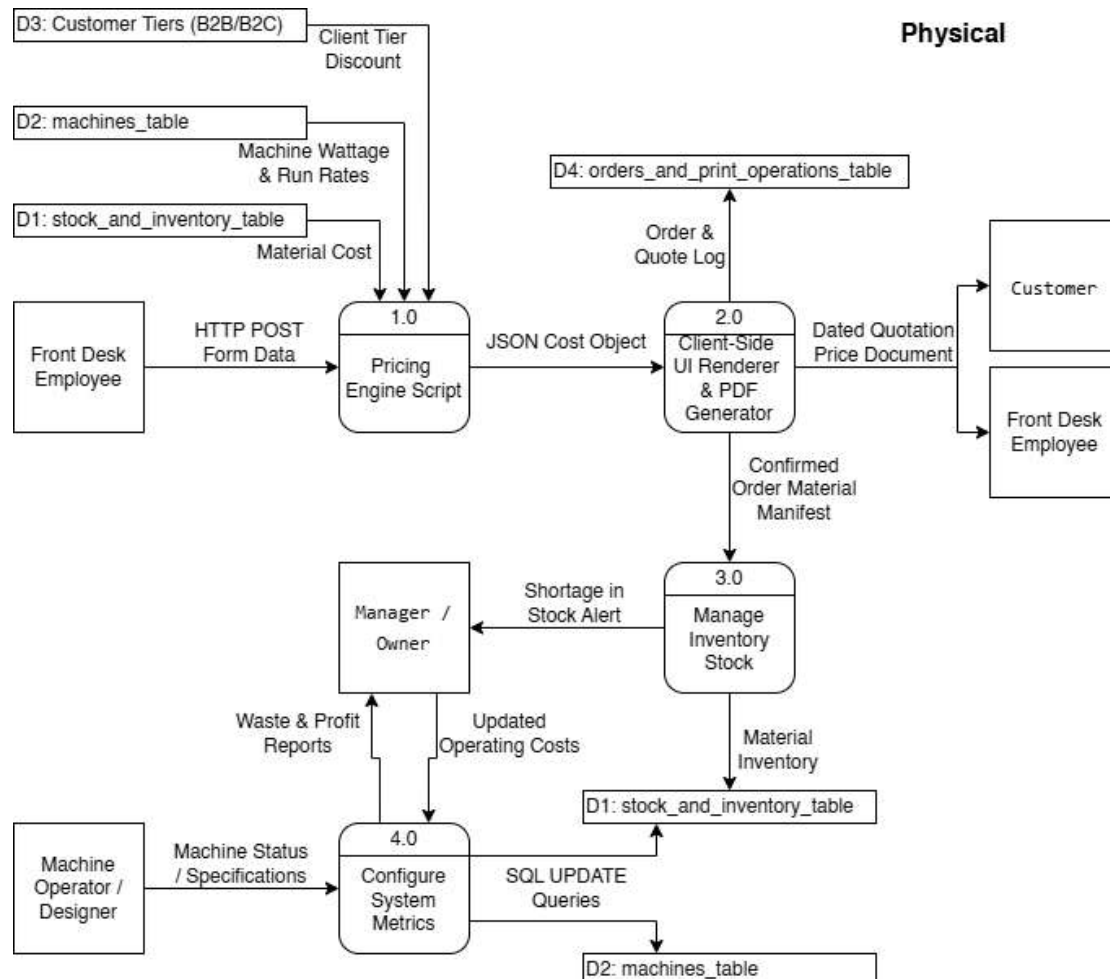
3.2. Logical Data Flow Diagram (Logical DFD)

The Logical DFD breaks down the system into core business processes (Calculate Order Cost, Generate Quotation, Manage Inventory). It demonstrates *what* the system does from a business perspective, ensuring all printing requirements, waste metrics, and customer tiers are logically routed to generate a final quotation and update material stock.



3.3. Physical Data Flow Diagram (Physical DFD)

Transitioning to implementation, the Physical DFD maps the business logic to actual technological constructs. It illustrates *how* the system operates, replacing logical stores with actual database tables (e.g., `stock_and_inventory_table`, `machines_table`) and detailing data payloads (e.g., HTTP POST Form Data, JSON Cost Object, SQL UPDATE Queries).



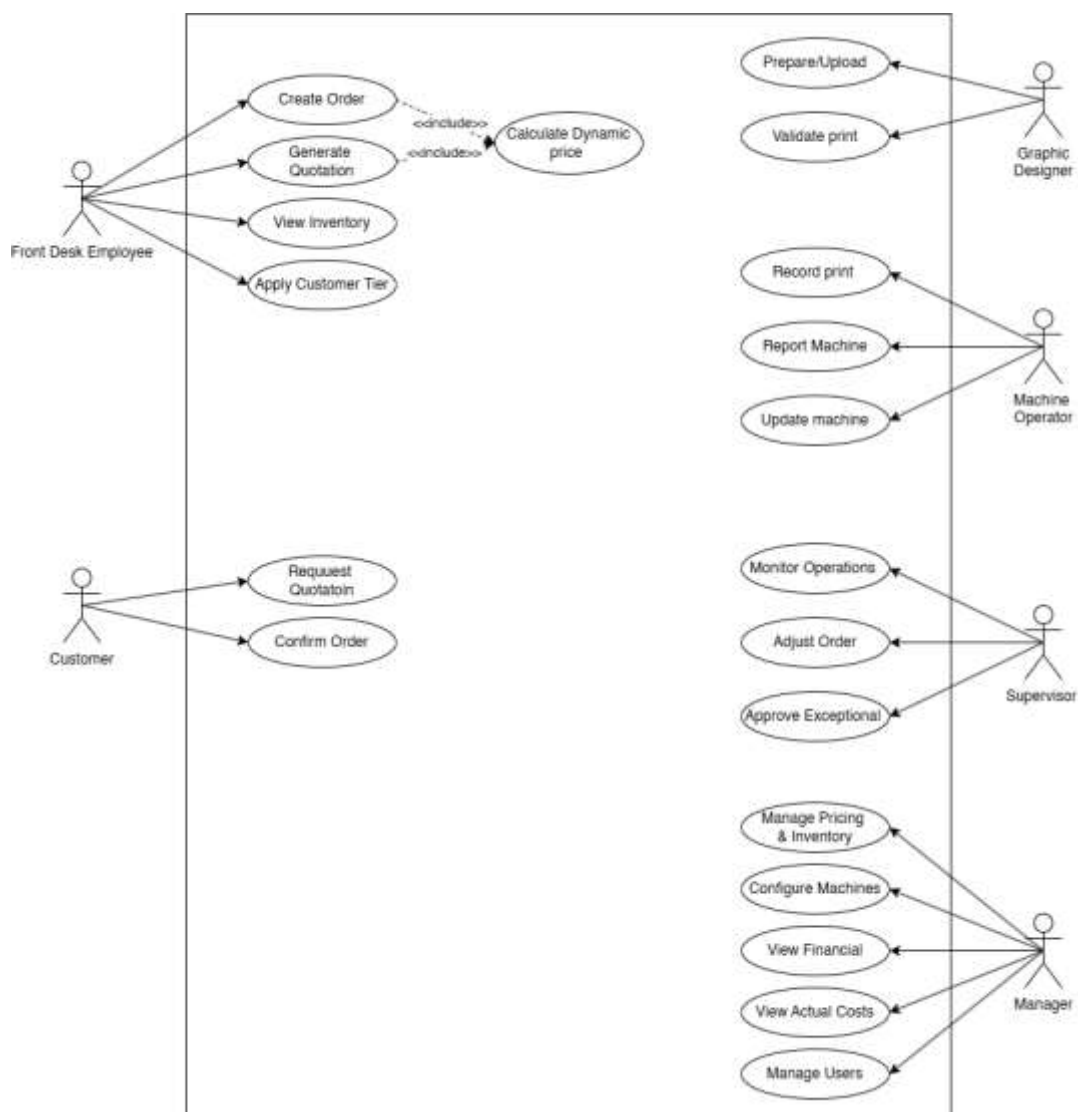
3.4. Use Case Diagram (Object-Oriented Behavioral Modeling)

In strict alignment with the Hybrid Methodology established in Phase 1, the system architecture bridges Structured Analysis with Object-Oriented (O-O) design principles.

While the Data Flow Diagrams (DFDs) capture the functional data transformations and storage mechanics, the Use Case Diagram models the system's behavioral requirements from an actor-centric perspective.

It explicitly maps out the operational boundaries and access privileges for each designated role (Front Desk Employee, Graphic Designer, Machine Operator, Supervisor, and Manager).

Furthermore, it highlights essential system dependencies using standard UML include relationships, ensuring that core algorithmic logic—such as dynamic pricing calculations—is uniformly invoked during both order entry and quotation generation sub-routines.



4. Event Response Table

The Event Response Table acts as the system's behavioral core. It dictates how the PrintSmart software will programmatically respond to specific real-world triggers inside the print shop, directly reflecting the processes modeled in the DFDs.

Table 4.1: PrintSmart Event Response Table

Source	Trigger	Activity	Response	Destination
Front Desk Employee	Order details (Material, size, colors) entered	Calculate Dynamic Price	Detailed Price Quotation	Front Desk Employee
Front Desk Employee	Payment confirmation & Order approval	Create Order & Generate Receipt	Order Receipt + Job Ticket	Customer + Machine Operator
Machine Operator	Job Completion + Actual waste logged	Record Print & Update Machine/Inventory	Production & Waste Report	Manager
Temporal (System Time)	Stock level reaches defined threshold	Manage Pricing & Inventory (Auto-scan)	Low Stock Alert	Manager